

KIRKCOFF UNIVERSITY KIRK 222 – Digital 1 page (Fall 2024) Final and Figure – Question 6!

Section 1: Basics

- 1. What is the primary purpose of a digital circuit?

☐ To store data permanently
 ☒ To perform logical operations on data
 ☐ To convert analog signals to digital
 ☐ To amplify signals
- 2. What is the basic unit of digital information?

☐ Bit
 ☒ Byte
 ☐ Kilobyte
 ☐ Megabyte
- 3. How many bits are in a byte?

☐ 8
 ☒ 16
 ☐ 32
 ☐ 64
- 4. What is the difference between a digital signal and an analog signal?

☐ Digital signals are continuous, while analog signals are discrete.
 ☒ Digital signals are discrete, while analog signals are continuous.
 ☐ Digital signals have a fixed amplitude, while analog signals vary.
 ☐ Digital signals are binary, while analog signals are multi-valued.

Section 2: Combinational Logic

Question 1: Simplify the expression

$$A + B + C + D + E + F + G + H + I + J + K + L + M + N + O + P + Q + R + S + T + U + V + W + X + Y + Z$$

☐ $A + B + C + D + E + F + G + H + I + J + K + L + M + N + O + P + Q + R + S + T + U + V + W + X + Y + Z$
☒ $A + B + C + D + E + F + G + H + I + J + K + L + M + N + O + P + Q + R + S + T + U + V + W + X + Y + Z$
☐ $A + B + C + D + E + F + G + H + I + J + K + L + M + N + O + P + Q + R + S + T + U + V + W + X + Y + Z$
☐ $A + B + C + D + E + F + G + H + I + J + K + L + M + N + O + P + Q + R + S + T + U + V + W + X + Y + Z$

Question 2: What is the output of the circuit?

☐ 0
☒ 1
☐ 2
☐ 3

Section 3: Sequential Logic

Question 1: What is the output of the circuit?

☐ 0
☒ 1
☐ 2
☐ 3

Question 2: What is the output of the circuit?

☐ 0
☒ 1
☐ 2
☐ 3

Section 4: Advanced Topics

Question 1: What is the output of the circuit?

☐ 0
☒ 1
☐ 2
☐ 3

Question 2: What is the output of the circuit?

☐ 0
☒ 1
☐ 2
☐ 3

Section 5: Final Review

Question 1: What is the output of the circuit?

☐ 0
☒ 1
☐ 2
☐ 3

Question 2: What is the output of the circuit?

☐ 0
☒ 1
☐ 2
☐ 3

[illegible][illegible]

Exercise 1

Figure 1 shows a logic circuit with two inputs, A and B, and one output, Y. The circuit consists of an AND gate followed by an OR gate.

```

graph LR
    A --> AND[AND]
    B --> AND
    AND --> OR[OR]
    B --> OR
    OR --> Y
  
```

The truth table for this circuit is shown below:

A	B	Y
0	0	0
0	1	1
1	0	0
1	1	1

Module 10: Statistics
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Question 1 (10%)

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Percent of respondents

x	y	x	xy
0	0	0	0
0	1	0	0
0	1	0	0
1	0	1	0
1	0	1	0
1	1	1	1
1	1	1	1

What is the least value, possibly, for the number of non-adjacent edges in a spanning tree?

For each value, explain how that is the number of non-adjacent edges in a spanning tree.

[illegible]

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Exercise 2 (10 points)

Exercise 2.1 (5 points)

Consider the following circuit:

```

graph LR
    x1((x1)) --- AND1[AND]
    x2((x2)) --- AND1
    x2 --- OR1[OR]
    x3((x3)) --- OR1
    AND1 --- OR2[OR]
    x1 --- OR2
    OR2 --- z((z))
    
```

Exercise 2.2 (5 points)

Consider the circuit representing the following function:

$$f(x_1, x_2, x_3) = (x_1 \vee x_2) \wedge (x_1 \vee x_3) \vee (x_2 \wedge x_3)$$

Draw the circuit representing the function f .

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Exercise 7: Adding

Consider the following circuit:

4-bit adder circuit diagram showing inputs x_3, x_2, x_1, x_0 and y_3, y_2, y_1, y_0 , and a carry-in z_0 . The circuit uses 4-bit registers and a 4-bit adder block. The output is a 4-bit sum f and a carry-out z_4 .

Truth table for the 4-bit adder:

x_3	x_2	x_1	x_0	y_3	y_2	y_1	y_0	z_0	f_3	f_2	f_1	f_0	z_4
0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	1	0	0	0	0	1	0
0	0	0	0	0	0	0	1	1	0	0	0	2	0
0	0	0	0	0	0	1	0	0	0	0	1	0	0
0	0	0	0	0	0	1	1	0	0	0	2	0	0
0	0	0	0	0	1	0	0	0	0	0	1	0	0
0	0	0	0	0	1	0	1	0	0	0	2	0	0
0	0	0	0	0	1	1	0	0	0	0	3	0	0
0	0	0	0	0	1	1	1	0	0	0	4	0	0
0	0	0	1	0	0	0	0	0	0	1	0	0	0
0	0	0	1	0	0	0	1	0	0	1	1	0	0
0	0	0	1	0	0	1	0	0	0	1	2	0	0
0	0	0	1	0	0	1	1	0	0	1	3	0	0
0	0	0	1	0	1	0	0	0	0	1	1	1	0
0	0	0	1	0	1	0	1	0	0	1	2	1	0
0	0	0	1	0	1	1	0	0	0	1	3	1	0
0	0	0	1	0	1	1	1	0	0	1	4	1	0
0	0	1	0	0	0	0	0	0	1	0	0	0	0
0	0	1	0	0	0	0	1	0	1	0	1	0	0
0	0	1	0	0	0	1	0	0	1	1	2	0	0
0	0	1	0	0	0	1	1	0	1	2	3	0	0
0	0	1	0	0	1	0	0	0	1	1	1	1	0
0	0	1	0	0	1	0	1	0	1	2	2	1	0
0	0	1	0	0	1	1	0	0	1	3	2	1	0
0	0	1	0	0	1	1	1	0	1	4	2	1	0
0	0	1	1	0	0	0	0	0	1	0	0	1	0
0	0	1	1	0	0	0	1	0	1	0	1	1	0
0	0	1	1	0	0	1	0	0	1	1	2	1	0
0	0	1	1	0	0	1	1	0	1	2	3	1	0
0	0	1	1	0	1	0	0	0	1	1	1	2	0
0	0	1	1	0	1	0	1	0	1	2	2	2	0
0	0	1	1	0	1	1	0	0	1	3	2	2	0
0	0	1	1	0	1	1	1	0	1	4	2	2	0
0	1	0	0	0	0	0	0	0	1	0	0	0	0
0	1	0	0	0	0	0	1	0	1	0	1	0	0
0	1	0	0	0	0	1	0	0	1	1	2	0	0
0	1	0	0	0	0	1	1	0	1	2	3	0	0
0	1	0	0	0	1	0	0	0	1	1	1	1	0
0	1	0	0	0	1	0	1	0	1	2	2	1	0
0	1	0	0	0	1	1	0	0	1	3	2	1	0
0	1	0	0	0	1	1	1	0	1	4	2	1	0
0	1	0	1	0	0	0	0	0	1	0	0</		